

Aditya Velpula

Fairfax, VA (Open to remote / NYC / SF / DMV / Seattle / Atlanta) | +1 571-609-8413 | reachaditya29@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | F1 STEM OPT Eligible

AI and Data Engineer who ships **production AI applications, scalable ETL pipelines, and the engineering rigor (1,481 automated tests, Langfuse observability, ~80% cost reduction via eval-driven LLM routing)** that keeps real systems trustworthy. Built + deployed Pulse, a public real-time AI platform with Claude streaming at [global-pulse-ai.site](#). Daily Python, advanced SQL, AWS production (S3, Glue, RDS, Lambda, Bedrock), LLMs (Claude, GPT-4, Gemini), RAG + embeddings, dimensional modeling, Power BI, Docker, GitHub Actions CI/CD. 5x AWS Certified incl. Gen AI Developer Pro + Data Engineer Associate. M.S. Data Analytics Engineering, GMU (May 2026).

RELEVANT EXPERIENCE

AI / Data Engineer (4-LLM Routing + ETL + Production AI), DAPSE Capstone | George Mason University

Jan 2026 to May 2026

NSI (National Security Innovations Inc.) Partner Organization

Arlington, VA

- Built a 7-stage async ETL pipeline with checkpoint recovery (AWS S3 + Glue, dimensional modeling, 1,481 automated quality tests in CI) over 1,630 multi-source records and shipped a 4-LLM routing layer (Claude / GPT-4 / Gemini / internal) with eval-driven selection, RAG + grounding, agentic MCP tool-use, Langfuse + Prometheus observability that cut compute + API spend ~80%.

Software / Data Engineer (Full-Stack) | Indgeos Geospatial

Nov 2023 to Jul 2024

Telangana, India

- Built Python + FastAPI services and SQL transformations against PostgreSQL backends, plus TypeScript / JavaScript frontends powering customer-facing geospatial analytics; ran early prompt-engineered LLM experiments on field-ops summaries.
- Stood up a Git-based pull-request workflow with linting, unit tests, and CI/CD checks, cutting code-review cycle time from ~3 days to under 24 hours across a 5-engineer team.

Graduate Teaching Assistant, AIT-580 Data Analytics | George Mason University

Aug 2025 to May 2026

Fairfax, VA

- Supported 100+ students for Prof. Harry Foxwell across Python, SQL, applied analytics, and data modeling; coached daily on translating business questions into structured data solutions.

PROJECTS

Pulse: Real-Time Global News Intelligence Globe with Claude-Powered Briefings | Next.js 16, React 19, TypeScript, Three.js + GLSL, Anthropic Claude API, Vercel | Live at [global-pulse-ai.site](#) | Apr 2026 to Present

- Solo-built public platform aggregating 1,000+ live events from 45+ sources (GDELT 2.0, 45 subreddits, RSS from Reuters / BBC / AP / NHK, Hacker News) with two-tier geocoding (200+ places + 75+ newsroom-HQ fallback) and grid-based spatial clustering on a 3D Earth (8K NASA textures, GLSL shaders for day/night terminator + sentiment-reactive auroras).
- Streams Claude Opus briefings via ReadableStream in ~1s (vs ~5s baseline); multi-language (es/fr/hi/zh/ar/de), fingerprint-based 10-min cache, per-IP rate limiting, Promise.allSettled with per-source timeouts, WebGL error boundary. Production-hardened (CSP, same-origin AI guards, spend caps).

Cloud Data Warehouse + ETL Pipeline on AWS (Public Health Risk Analytics) | AWS S3, Glue, RDS (PostgreSQL), PySpark, Python, R, SQL | Supervised by Prof. Foxwell

- End-to-end ETL on CDC BRFSS data with PySpark over raw / clean / curated layers (medallion architecture); dimensional modeling, automated data quality checks, R statistical analysis layer.

Predictive Analytics + BI Dashboard (SLA Breach Forecasting) | Python, XGBoost, Scikit-learn, advanced SQL, Power BI, SHAP | 2025

- Dimensional model + canonical dataset over 5,000+ ops records, XGBoost forecasting with SHAP explainability, Power BI dashboard with drill-down for operator self-serve.

RAG + Agentic AI Assistant with Citation Grounding + MCP Tool Use | Python, OpenAI API, Claude, Hugging Face, FAISS + pgvector, MCP servers, FastAPI | 2026

- Embedded structured + unstructured operational records, served retrieval through FAISS + pgvector, grounded LLM answers with citations, exposed analytics tools to the agent via MCP with guarded loops and token budgets.

Multi-Cloud LLM Evaluation Harness (Bedrock + OpenAI + Azure OpenAI + Vertex AI) | Python, AWS Bedrock, OpenAI, Azure OpenAI, Vertex AI, Pytest, Langfuse | 2026

- Cross-provider eval harness comparing Claude / GPT / Gemini on a fixed prompt + gold-answer set; powers model-selection decisions for the production 4-LLM router.

Real-Time Streaming Anomaly Detection on AWS (Sensor + Event Telemetry) | AWS Kinesis Data Streams, Lambda, DynamoDB, Python, Isolation Forest, LSTM autoencoder | 2025

- Ingested high-frequency event streams through Kinesis into Lambda consumers, scored each event with an Isolation Forest + LSTM-autoencoder ensemble, persisted alerts to DynamoDB; sub-second p95 latency end-to-end.

NLP Sentiment + Topic Classifier with Transformer Fine-Tuning | Python, PyTorch, Hugging Face Transformers (BERT / DistilBERT), Datasets, scikit-learn, FastAPI | 2025

- Fine-tuned DistilBERT on 50K+ labeled reviews for multi-class sentiment + topic tagging; served via FastAPI with batched inference and ONNX runtime acceleration for ~3x throughput.

Time-Series Demand Forecasting + Inventory Optimization | Python, Prophet, LightGBM, statsmodels (SARIMA), Optuna, MLflow, Power BI | 2024 to 2025

- Hierarchical forecasting across product / region with Prophet + LightGBM stacked ensemble, Optuna hyperparameter search tracked in MLflow; surfaced reorder recommendations through a Power BI ops dashboard.

End-to-End MLOps Pipeline (Training + Registry + Drift Monitoring) | Python, MLflow, AWS SageMaker, Docker, GitHub Actions, Evidently AI, Prometheus | 2026

- CI/CD pipeline that trains, evaluates, and registers models in MLflow + SageMaker Model Registry, with Evidently AI drift detection + Prometheus alerts wired into a re-training trigger when input distributions shift.

EDUCATION

George Mason University, Master of Science in Data Analytics Engineering Aug 2024 to May 2026 | Fairfax, VA

St. Peters Engineering College, Bachelor of Technology in Computer Science & Engineering (AI & ML) Aug 2020 to Mar 2024 | Telangana, India

TECHNICAL SKILLS

Programming & SQL: Python (5+ yrs daily, Pandas, NumPy, SQLAlchemy, FastAPI, OOP), advanced SQL (PostgreSQL, joins, window functions, CTEs, query optimization), TypeScript / JavaScript (production via Indgeos + Pulse), Bash, R.

AI / LLM & ML: Anthropic Claude API + streaming (Pulse production), OpenAI GPT-4/4o, Gemini; prompt engineering, RAG (FAISS + pgvector + hybrid retrieval), embeddings, agentic tool-use over MCP, LLM-as-judge evaluation. Scikit-learn, PyTorch, Hugging Face Transformers, XGBoost, LightGBM, Prophet; classical ML, NLP, SHAP/LIME.

Data Engineering & ETL: ETL/ELT pipelines, dimensional modeling (star/snowflake, SCD), data warehousing, PySpark distributed transforms, dbt patterns + Airflow/Prefect (familiar), Kafka / AWS Kinesis streaming, async orchestration with checkpoint recovery, schema enforcement + drift detection.

Cloud (AWS 5x Certified): AWS daily (S3, Glue, RDS, Lambda, Athena, EMR, Bedrock, Kinesis, SageMaker, DynamoDB, IAM, CloudWatch). AWS Certs: Gen AI Developer (Pro) · ML Engineer · Data Engineer · Solutions Architect · AI Practitioner [\[verify\]](#). Azure (Azure OpenAI), GCP / Vertex AI (awareness).

Data Quality, Observability, Viz, DevOps & Web: 1,481 automated CI tests (Great Expectations / Pydeequ patterns), schema validation, drift + freshness monitoring, Langfuse + Prometheus + Evidently AI. Power BI production, Tableau / Streamlit. Git + GitHub Actions CI/CD daily, Docker, Kubernetes familiar, pytest, MLflow, SageMaker Model Registry. Next.js 16, React 19, TypeScript, Three.js + WebGL + GLSL (Pulse), Vercel. Cursor + Claude AI-augmented dev.